Exercise 1

a) Accumulation points $= \{-1, 0, 1\}$

$\limsup = 1 \qquad \liminf = -1$

b) Acc. pts $= \{-\infty, +\infty\}$

$\limsup = +\infty \qquad \liminf = -\infty$

c) Accumpoints $= \{\frac{1}{2}, 2\}$

$\limsup = 2 \qquad \liminf = \frac{1}{2}$

d) Acc. pts $= [-1, 1]$

$\limsup = 1 \qquad \liminf = -1$

We had a change in the sheet Friday night!
For a few hours the exercise asked for $X = \{0, +\infty\}$
Points will be awarded for solutions to that one as well
(ex: $x_n = 0, 0, 0, 1, 0, 2, 0, 3, 0, 4, \dots$)

e) $(x_n)_n = \underbrace{1, 2, 3, 4}, \underbrace{1, 2, 3, 4}, \underbrace{1, 2, 3, 4}, \dots$

(1, 2, 3, 4 is repeated periodically)

f) $(x_n)_n = \underbrace{1, 2, 0}, \underbrace{1, 2, -1}, \dots, \underbrace{1, 2, -k}, \dots$

(block of 3-terms. The k-th being $1, 2, -k$)

g) $(x_n)_n = \underbrace{1}, \underbrace{1, 1/2}, \underbrace{1, 1/2, 1/3}, \underbrace{1, 1/2, 1/3, 1/4}, \dots$

(blocks of increasing length. The k-th being of length k given by $1, 1/2, 1/3, \dots, 1/(k-1), 1/k$)

Exercise 2

a) $x_n = 0 \le 6$ so base step is done.
Suppose $x_n \le 6$. Then $x_{n+1} = \sqrt{30 + x_n} \le \sqrt{30+6} \le 6$
And $x_k \le 6 \ \forall k \ge 0$ by induction

b) $0 \le x_n \le 6 \quad \forall n \ge 0$ (by def and by part a))

$x_{n+1} = \sqrt{30 + x_n} \overset{?}{\ge} x_n$

We solve $\sqrt{30 + x} \ge x$. If $x > 0$ it's equivalent to

$30 + x > x^2 \iff x^2 - x - 30 \le 0 \iff (x+5)(x-6) < 0$

True for $0 \le x \le 6$ as wanted.

c) x_n is bounded above by 6 and increasing $\Rightarrow x_n$ converges

Let $\lim_{n\to\infty} x_n = \ell$. Then also $\lim_{n\to\infty} x_{n+1} = \ell$. ~~For n→∞~~

We have $x_{n+1} = \sqrt{30 + x_n} \Rightarrow x_{n+1}^2 = 30 + x_n$ taking the limit and (using the product rule for the left side) $\ell^2 = 30 + \ell \Rightarrow \ell = 6, -5$ but -5 is not possible since $x_n \ge 0 \Rightarrow \boxed{\ell = 6}$

Exercise 3

a) x_{10k} is a subsequence of x_{2k} and x_{5k}. These latter two are convergent, and therefore x_{10k} has to converge to both their limits, hence x_{2k} and x_{5k} must have the same limit.

x_{10k+5} is a subsequence of x_{2k+1} and x_{5k}, and as above those last two must converge to the same value.
It follows that x_{2k} and x_{2k+1} have the same limit as well.

[The idea was to divide x_{5k} in two subsequences, the one with odd indices and the one with even ones and then deduce these two were convergent since x_{2k} and x_{2k+1} were]

b) Call the common limit of x_{2k}, x_{2k+1} ℓ. Fix $\varepsilon > 0$.
$\exists\ \bar{k}$ s.t. $\forall\ k > \bar{k}$ $|x_{2k} - \ell| < \varepsilon$, and also
$\exists\ \bar{\bar{k}}$ s.t. $\forall\ k > \bar{\bar{k}}$ $|x_{2k+1} - \ell| < \varepsilon$.
Then for all $n > 2\max\{\bar{k}, \bar{\bar{k}}\} + 1 := \bar{n}$ $|x_n - \ell| < \varepsilon$
Hence x_n converges to ℓ.
[We use that x_{2k} and x_{2k+1} is a partition of x_n].

Exercise 4

a) For $n = 0$ it's just $|x_1 - x_0| \leq \lambda^0 |x_1 - x_0|$ wich is true.
Suppose $|x_{n+1} - x_n| \leq \lambda^n |x_1 - x_0|$ then
$|x_{n+2} - x_{n+1}| \leq \lambda |x_{n+1} - x_n| \leq \lambda \cdot \lambda^n |x_1 - x_0| \leq \lambda^{n+1} |x_1 - x_0|$
as desired, hence the thesis is proven by induction.

b)
$$
\begin{aligned}
|x_m - x_n| &= |(x_m - x_{m-1}) + \dots + (x_{n+1} - x_n)| \\
&\leq |x_{n+1} - x_n| + \dots + |x_m - x_{m-1}| \\
&\leq \lambda^n |x_0 - x_1| + \dots + \lambda^{m-1} |x_0 - x_1| \\
&\leq \lambda^n \left(\frac{1 - \lambda^m}{1 - \lambda}\right) |x_0 - x_1| \leq \lambda^n \cdot \frac{1}{1-\lambda} |x_0 - x_1| := \lambda^n C
\end{aligned}
$$

doesn't depend on n.

If we fix $\varepsilon > 0$, then $\exists\ \bar{n}$ s.t. $\lambda^{\bar{n}} \cdot C < \varepsilon$
Then $\forall\ n, m > \bar{n}$
$|x_m - x_n| \leq \lambda^n \cdot C < \varepsilon \Rightarrow x_n$ is a Cauchy sequence
$\Rightarrow x_n$ is convergent

Exercise 5

a) We prove by induction that if $x_{\bar{n}}, x_{\bar{n}+1} \leq C$ then $x_n, x_{n+1} \leq C \quad \forall\, n \geq \bar{n}$.

The base step is trivial for $n = \bar{n}$.

Then suppose that $x_n, x_{n+1} \leq C$.

$x_{n+2} \leq \frac{x_n + x_{n+1}}{2} \leq \frac{C+C}{2} \leq C$ and we already know $x_{n+1} \leq C$.

Hence the induction is completed.

Then the sequence is bounded above by $\max\{x_0, x_1\}$ and below by 0.

Moreover if $x_{\bar{n}}, x_{\bar{n}+1} \leq C$, $x_n \leq C \quad \forall\, n > \bar{n} \Rightarrow \limsup_{n\to\infty} x_n \leq C$.

b) Fix $\varepsilon > 0$.

By def of limsup $\exists\, \bar{n}$ s.t. $\forall\, n \geq \bar{n} \quad x_n < L + \varepsilon$.

By characterization of liminf $\exists\, \tilde{n} > \bar{n}$ s.t. $x_{\tilde{n}} < \ell + \varepsilon$.

Notice that $\tilde{n} - 1 \geq \bar{n} \Rightarrow x_{\tilde{n}-1} < L + \varepsilon$.

Then $x_{\tilde{n}+1} \leq \frac{x_{\tilde{n}} + x_{\tilde{n}-1}}{2} \leq \frac{L+\varepsilon + \ell+\varepsilon}{2} \leq \frac{L+\ell}{2} + \varepsilon$

Applying a) to $x_{\tilde{n}}, x_{\tilde{n}+1} \leq \frac{L+\ell}{2} + \varepsilon$ (since $\ell + \varepsilon \leq \frac{L+\ell}{2} + \varepsilon$)

We get $\limsup x_n \leq \frac{L+\ell}{2} + \varepsilon$. Since ~~this~~ holds for every ε greater than 0 we get $L \leq \frac{L+\ell}{2}$.

c) From b) we know $L \leq \frac{L+\ell}{2}$, but since $\ell \leq L$ we have $L \leq \frac{L+\ell}{2} \leq L$. Then both inequalities must be equalities.

And then $L = \frac{L+\ell}{2} \Rightarrow \frac{L}{2} = \frac{\ell}{2} \Rightarrow L = \ell$

Since $\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n$ x_n must converge.